HW 3 due 3/3

1) b) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \iff x_1+x_2+x_3+x_4 = 0$

$x_2+2x_2+3x_3+4x_4 = 0$

a) By b) and proposition 14.3, since V is exactly the kernel of the linear transformation $T\vec{x} = A\vec{x}$, and ker(A) is a subspace of $\mathbb{R}^4$, V is a subspace of $\mathbb{R}^4$ too.

c) B s.t. im(B) = V.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 \end{array}\right] \xrightarrow{r_2 - r_1} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{array}\right] \xrightarrow{r_1 - r_2} \left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{array}\right]$$

$x_1 - x_3 - 2x_4 = 0$ $\quad x_3 = s, x_4 = t$

$x_2 + 2x_3 + 3x_4 = 0$

$x_1 = s + 2t$

$x_2 = -2s - 3t$

$$\ker(A) = \left\{ \begin{pmatrix} s+2t \\ -2s-3t \\ s \\ t \end{pmatrix}, s,t \in \mathbb{R} \right\}$$

$$\text{basis for } \ker(A) = V = \text{im}(B) = \left\{ s\begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix}, s,t \in \mathbb{R} \right\}$$

$$B = \begin{pmatrix} 1 & 2 \\ -2 & -3 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2) a) $|\vec{v}| = \sqrt{(-3)^2+1^2} = \sqrt{10}$ $\quad |\vec{w}| = \sqrt{(-2)^2+(-1)^2} = \sqrt{5}$

$\vec{v} \cdot \vec{w} = (-3)(-2) + (1)(-1) = 5$

$\angle(\vec{v}, \vec{w}) = \arccos\left(\frac{5}{\sqrt{10}\sqrt{5}}\right) = \arccos\left(\frac{5}{\sqrt{50}}\right) = \arccos\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$ acute

b/c $0 < \frac{1}{\sqrt{2}} < 1$

Area:

$\sqrt{|\vec{v}|^2|\vec{w}|^2 - |\vec{v}\cdot\vec{w}|^2} = \sqrt{10 \cdot 5 - 5^2} = \sqrt{50-25} = 5$

b) $|\vec{v}| = \sqrt{2^2+3^2+4^2} = \sqrt{29}$ $\quad |\vec{w}| = \sqrt{2^2+(-8)^2+5^2} = \sqrt{93}$ $\quad \vec{v}\cdot\vec{w} = (2\cdot 2)+(3\cdot -8)+(4\cdot 5) = 0$

$\angle(\vec{v}, \vec{w}) = \arccos(0) = \frac{\pi}{2}$ right angle since $\vec{v}\cdot\vec{w} = 0$

Area:

$\sqrt{29 \cdot 93 - 0} = \sqrt{2697}$

3) $\vec{v_1} = (4,0,3)$ $\vec{v_2} = (25,0,25)$ $\vec{v_3} = (0,-2,0)$

$\vec{u_1} = \frac{\vec{v_1}}{|\vec{v_1}|} = \frac{(4,0,3)}{\sqrt{25}} = \left(\frac{4}{5}, 0, \frac{3}{5}\right)$

$\text{proj}_{S_1}\vec{v_2} = (\vec{v_2}\cdot\vec{u_1})\vec{u_1} = (35)\left(\frac{4}{5},0,\frac{3}{5}\right) = (28,0,21)$

$\text{perp}_{S_1}\vec{v_2} = \vec{v_2} - \text{proj}_{S_1}\vec{v_2} = (25,0,25) - (28,0,21) = (-3,0,4)$

$\vec{u_2} = \frac{\text{perp}_{S_1}\vec{v_2}}{|\text{perp}_{S_1}\vec{v_2}|} = \frac{(-3,0,4)}{\sqrt{25}} = \left(-\frac{3}{5}, 0, \frac{4}{5}\right)$

$\text{proj}_{S_2}\vec{v_3} = (\vec{v_3}\cdot\vec{u_1})\vec{u_1} + (\vec{v_3}\cdot\vec{u_2})\vec{u_2}$

$= 0 + 0$

$= 0$

$\text{perp}_{S_2}\vec{v_3} = \vec{v_3} - \text{proj}_{S_2}\vec{v_3} = (0,-2,0)$

$\vec{u_3} = \frac{(0,-2,0)}{\sqrt{4}} = (0,-1,0)$

$$\left\{\begin{bmatrix}\frac{4}{5}\\0\\\frac{3}{5}\end{bmatrix}, \begin{bmatrix}-3/5\\0\\4/5\end{bmatrix}, \begin{bmatrix}0\\-1\\0\end{bmatrix}\right\}$$

4) basis of ker(A) (from HW 2):

$$\left\{\begin{bmatrix}1/2\\-1/2\\1\\0\end{bmatrix}, \begin{bmatrix}5/6\\-1/6\\0\\1\end{bmatrix}\right\} \quad \vec{v_1} \quad \vec{v_2}$$

$\vec{u_1} = \frac{\vec{v_1}}{\sqrt{2}} = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, 0\right)$

$\text{proj}_{S_1}\vec{v_2} = (\vec{v_2}\cdot\vec{u_1})\vec{u_1} = \left(\frac{5}{\sqrt{36}} + \frac{1}{\sqrt{36}} + 0 + 0\right)\vec{u_1} = \frac{1}{\sqrt{6}}\vec{u_1} = \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, 0\right)$

$\text{perp}_{S_1}\vec{v_2} = \vec{v_2} - \text{proj}_{S_1}\vec{v_2} = \left(\frac{5}{6}, -\frac{1}{6}, 0, 1\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, 0\right) = \left(\frac{4}{3}, 0, -\frac{1}{3}, 1\right)$

$\vec{u_2} = \frac{\left(\frac{2}{3}, 0, -\frac{1}{3}, 1\right)}{\sqrt{\frac{4}{9} + 0 + \frac{1}{9} + 1}} = \frac{3\left(\frac{2}{3}, 0, -\frac{1}{3}, 1\right)}{\sqrt{14}} = \left(\frac{2}{\sqrt{14}}, 0, -\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$

$$\left\{\begin{bmatrix}1/\sqrt{6}\\-1/\sqrt{6}\\\sqrt{2}/\sqrt{3}\\0\end{bmatrix}, \begin{bmatrix}2/\sqrt{14}\\0\\-1/\sqrt{14}\\3/\sqrt{14}\end{bmatrix}\right\} = \left\{\begin{bmatrix}\frac{\sqrt{6}}{6}\\-\frac{\sqrt{6}}{6}\\\frac{\sqrt{6}}{3}\\0\end{bmatrix}, \begin{bmatrix}\frac{\sqrt{14}}{7}\\0\\-\frac{\sqrt{14}}{14}\\\frac{3\sqrt{14}}{14}\end{bmatrix}\right\}$$

4) cont. basis of im(T) from HW 2: $\{\vec{v}_1, \vec{v}_2\} = \left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}5\\11\\6\end{bmatrix}\right\}$

$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{(1,1,0)}{\sqrt{2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$

$\text{proj}_{\vec{u}_1}\vec{v}_2 = (\vec{v}_2 \cdot \vec{u}_1)\vec{u}_1 = \left(\frac{5}{\sqrt{2}} + \frac{11}{\sqrt{2}} + 0\right)\vec{u}_1 = \left(\frac{16}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) = (8, 8, 0)$

$\text{perp}_{\vec{u}_1}\vec{v}_2 = \vec{v}_2 - (8,8,0) = (5,11,6) - (8,8,0) = (-3, 3, 6)$

$\vec{u}_2 = \frac{(-3,3,6)}{\sqrt{54}} = \frac{(-3,3,6)}{3\sqrt{6}} = \left(\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$

$$\left\{\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}, \begin{bmatrix}-1/\sqrt{6}\\1/\sqrt{6}\\2/\sqrt{6}\end{bmatrix}\right\} = \left\{\begin{bmatrix}2/\sqrt{2}\\2/\sqrt{2}\\0\end{bmatrix}, \begin{bmatrix}-\sqrt{6}/6\\\sqrt{6}/6\\\sqrt{6}/3\end{bmatrix}\right\}$$

5) ① $A = \begin{bmatrix}a & b & 0\\ c & d & 0\\ e & f & 1\end{bmatrix}$ $A^T = \begin{bmatrix}a & c & e\\ b & d & f\\ 0 & 0 & 1\end{bmatrix}$ A orthogonal $= AA^T = I \Rightarrow A^TA = I$

$e = f = 0$

② $A^TA = \begin{bmatrix}a & c & e\\ b & d & f\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}a & b & 0\\ c & d & 0\\ e & f & 1\end{bmatrix} = \begin{bmatrix}a^2+c^2+e^2 & ab+cd+ef & e\\ ab+cd+ef & b^2+d^2+f^2 & f\\ e & f & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

③ substitute $e=f=0$ (1) $a^2+c^2=1$

$\begin{bmatrix}a^2+c^2 & ab+cd & 0\\ ab+cd & b^2+d^2 & 0\\ 0 & 0 & 1\end{bmatrix}$ (2) $b^2+d^2=1$

(3) $ab+cd=0 \Rightarrow (ab)^2=(-cd)^2$

$a^2b^2 = c^2d^2$

④ ① $\times b^2$: $a^2b^2 + b^2c^2 = b^2$

② $\times c^2$: $b^2c^2 + \underbrace{c^2d^2}_{\text{substitute}} = c^2$

$b^2c^2 + a^2b^2 = c^2$

equivalent, so $b^2 = c^2$

$b = \pm c$

⑤ case 1: $b=c$ $\quad ac+cd=0$, $c \neq 0$, $a+d=0$, $a=-d$

case 2: $b=-c$ $\quad -ac+cd=0$, $-a+d=0$, $a=d$

⑥ $b=c$, $a=-d$: $\begin{bmatrix}a & c & 0\\ c & -a & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}\cos\theta & \sin\theta & 0\\ \sin\theta & -\cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$

$b=-c$, $a=d$: $\begin{bmatrix}a & -c & 0\\ c & a & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$

$\frac{15}{3}$ $\frac{8}{3}$ $\frac{9}{3} - \frac{23}{3}$

6) a) $\left[\begin{array}{cc|c} 3 & 2 & 5 \\ 5 & 3 & 9 \\ 4 & 5 & 3 \end{array}\right] \xrightarrow{r_1(\frac{1}{3})} \left[\begin{array}{cc|c} 1 & 2/3 & 5/3 \\ 5 & 3 & 9 \\ 4 & 5 & 3 \end{array}\right] \xrightarrow[r_3 - 4r_1]{r_2 - 5r_1} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{5}{3} \\ 0 & -1/3 & 2/3 \\ 0 & -7/3 & -16/3 \end{array}\right]$

$\xrightarrow{r_1 + 2r_2} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & -1/3 & 2/3 \\ 0 & -7/3 & -16/3 \end{array}\right] \xrightarrow[r_3 \cdot (3)]{r_2 \cdot -(3)} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 7 & -16 \end{array}\right] \xrightarrow{r_3 - 7(r_2)} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{array}\right]$

$0 \neq -2$ so $A\vec{x} = \vec{b}$ no sol

b) $\vec{x} = (A^T A)^{-1} A^T \vec{b}$

① $A^T A = \begin{bmatrix} 3 & 5 & 4 \\ 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 9+25+16 & 6+15+20 \\ 6+15+20 & 4+9+25 \end{bmatrix} = \begin{bmatrix} 50 & 41 \\ 41 & 38 \end{bmatrix}$

② $(A^T A)^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{(50 \cdot 38) - (41)^2} \begin{bmatrix} 38 & -41 \\ -41 & 50 \end{bmatrix} = \frac{1}{219} \begin{bmatrix} 38 & -41 \\ -41 & 50 \end{bmatrix}$

$1900 - 1681$

③ $(A^T A)^{-1} A^T = \frac{1}{219} \begin{bmatrix} 38 & -41 \\ -41 & 50 \end{bmatrix} \begin{bmatrix} 3 & 5 & 4 \\ 2 & 3 & 5 \end{bmatrix} = \frac{1}{219} \begin{bmatrix} 114-82 & 190-123 & 152-205 \\ -123+100 & -205+150 & -164+250 \end{bmatrix} =$

$\star \vec{b}$ ④ $\frac{1}{219} \begin{bmatrix} 32 & 67 & -53 \\ -23 & -55 & 86 \end{bmatrix} \begin{bmatrix} 5 \\ 9 \\ 3 \end{bmatrix} = \frac{1}{219} \begin{bmatrix} 160+603-159 \\ -115-495+258 \end{bmatrix} = \frac{1}{219} \begin{bmatrix} 604 \\ -352 \end{bmatrix} =$

$2 \times 3$ $3 \times 1$

$\vec{x} = \begin{bmatrix} 604/219 \\ -352/219 \end{bmatrix}$

c) $A\vec{x} - \vec{b} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} \frac{604}{219} \\ \frac{-352}{219} \end{bmatrix} - \begin{bmatrix} 5 \\ 9 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{1812}{219} - \frac{704}{219} \\ \frac{3020 - 1056}{219} \\ \frac{2416 - 1760}{219} \end{bmatrix} - \begin{bmatrix} 5 \\ 9 \\ 3 \end{bmatrix} =$

$\begin{bmatrix} 1108/219 \\ 1964/219 \\ 656/219 \end{bmatrix} - \begin{bmatrix} 5 \\ 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 13/219 \\ -7/219 \\ -1/219 \end{bmatrix}$

$|A\vec{x} - \vec{b}| = \sqrt{\left(\frac{13}{219}\right)^2 + \left(\frac{-7}{219}\right)^2 + \left(\frac{-1}{219}\right)^2} = \sqrt{\frac{169+49+1}{2192}} = \frac{1}{\sqrt{219}}$

$|\vec{b}| = \sqrt{25 + 81 + 9} = \sqrt{115}$

$\frac{1}{\sqrt{219}} \cdot \frac{1}{\sqrt{115}} = \frac{1}{\sqrt{25185}} \approx 0.006301$ $\approx 0.63\%$

7) $r(N) = k/t$, let $k = 1$ $\quad r(N) = 1/t$

$r(N) = aN + b$

$r(1) = \frac{1}{58} = a + b$

$r(2) = \frac{1}{31} = 2a + b$

$r(3) = \frac{1}{19} = 3a + b$

$r(4) = \frac{1}{15} = 4a + b$

$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}$ $\quad \vec{x} = \begin{bmatrix} a \\ b \end{bmatrix}$ $\quad \vec{b} = \begin{bmatrix} \frac{1}{58} \\ \frac{1}{31} \\ \frac{1}{19} \\ \frac{1}{15} \end{bmatrix}$ $\quad A^T = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

$A^TA = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}$ $\quad (A^TA)^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix}$

$A^Tb = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/58 \\ 1/31 \\ 1/19 \\ 1/15 \end{bmatrix} = \begin{bmatrix} \frac{1}{58} + \frac{2}{31} + \frac{3}{19} + \frac{4}{15} \\ \frac{1}{58} + \frac{1}{31} + \frac{1}{19} + \frac{1}{15} \end{bmatrix} = \begin{bmatrix} 0.5063189 \\ 0.1687977 \end{bmatrix}$

$\vec{x} = (A^TA)^{-1}A^T\vec{b} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 0.5063189 \\ 0.1687977 \end{bmatrix} = \begin{bmatrix} 1/5 & -1/2 \\ -1/2 & 3/2 \end{bmatrix} \begin{bmatrix} 0.5063189 \\ 0.1687977 \end{bmatrix} = \begin{bmatrix} 0.01686 \\ 0.0000371 \end{bmatrix}$ (a, b)

$r(15) = (0.01686)(15) + (0.0000371) = 0.2529371 \approx 0.253$

$1/0.253 = 3.953$ min

$$\vec{v_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \vec{v_2} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

8) trying to find proj span $(\vec{v_1}, \vec{v_2})(\vec{x})$ =

Let $V = \text{span}(\vec{v_1}, \vec{v_2})$ ; Let $\vec{u_1} \ldots \vec{u_m}$ be an orthonormal basis for V

$\text{proj}_V \vec{x} = (x \cdot \vec{u_1})u_1 + \cdots (x \cdot \vec{u_m})u_m$

$\vec{x_1} = \vec{v_1}$

$\vec{x_2} = \vec{v_2} - \text{proj}_{\vec{x_1}}(\vec{v_2})$

$= \vec{v_2} - \left(\frac{\vec{x_1} \cdot \vec{v_2}}{\|\vec{x_1}\|^2}\right)\vec{x_1}$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \left(\frac{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}}{1}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\vec{u_2} = \frac{\vec{x_2}}{|\vec{x_2}|} = \frac{\begin{bmatrix} 0 \\ 2 \\ 3 \\ 4 \end{bmatrix}}{\sqrt{29}} = \begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} \qquad \vec{u_1} = \vec{v_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ (unit vector sum)}$$

$P(\vec{x}) = \text{proj}_V \vec{x} = (\vec{u_1} \cdot x)\vec{u_1} + (\vec{u_2} \cdot x)\vec{u_2}$

$$P(e_1) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \left(\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + 0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$P(e_2) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \left(\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = 0 + (2/\sqrt{29})\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = \begin{bmatrix} 0 \\ 4/29 \\ 6/29 \\ 8/29 \end{bmatrix}$$

$$P(e_3) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \left(\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}\right)\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = 0 + (3/\sqrt{29})\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = \begin{bmatrix} 0 \\ 6/29 \\ 9/29 \\ 12/29 \end{bmatrix}$$

$$P(e_4) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \left(\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right)\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = 0 + (4/\sqrt{29})\begin{bmatrix} 0 \\ 2/\sqrt{29} \\ 3/\sqrt{29} \\ 4/\sqrt{29} \end{bmatrix} = \begin{bmatrix} 0 \\ 8/29 \\ 12/29 \\ 16/29 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4/29 & 0 & 0 \\ 0 & 0 & 9/29 & 0 \\ 0 & 0 & 0 & 16/29 \end{bmatrix} \qquad P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4/29 & 6/29 & 8/29 \\ 0 & 6/29 & 9/29 & 12/29 \\ 0 & 8/29 & 12/29 & 16/29 \end{bmatrix}$$